

Explanation of Approach for Complex Queries

Query 5: Book Popularity Report

Objective:

Identify books that have been loaned at least three times and display their borrowing frequency along with average review ratings.

Approach:

This query joins the **Book** table with the **Loan** table to count how many times each book has been borrowed. A `GROUP BY` clause is used on book attributes to aggregate loan counts. To ensure only popular books are included, the `HAVING COUNT(LoanID) >= 3` condition is applied after grouping.

To include review information, a `LEFT JOIN` is used with the **Review** table. This ensures that books without reviews are still displayed while allowing the calculation of the average rating using `AVG(Rating)`. Using `LEFT JOIN` avoids excluding books that meet the loan requirement but have no reviews.

Query 6: Member Reading History

Objective:

Display the complete borrowing history for each member, including current and past loans, along with any reviews submitted.

Approach:

This query starts from the **Members** table and joins it with the **Loan** and **Book** tables to retrieve all borrowing records for each member. Both active and returned loans are included to ensure the history is complete.

A `LEFT JOIN` is used with the **Review** table to attach review information where available. This design allows members who did not leave reviews to still appear in the result set. Loan dates and return dates are selected directly from the loan records to reflect borrowing activity accurately. The result can be ordered by loan date to present the reading history in a logical timeline.

Query 7: Revenue Analysis by Genre

Objective:

Calculate total fines collected per book genre and analyze loan and payment patterns.

Approach:

This query joins the **Book**, **Loan**, and **Payment** tables to associate fine payments with the corresponding book genre. The query groups results by genre to calculate aggregate values such as total loans, total fine amount collected, and average fine per loan.

Aggregate functions such as `COUNT()`, `SUM()`, and `AVG()` are used to produce financial insights. `ISNULL` (or `COALESCE`) is applied where necessary to handle loans that do not have associated fines, ensuring accurate calculations and preventing `NULL` values from affecting totals.

Screenshots of query results for at least 5 different queries

```
--2. Active Borrowers Analysis
select M.FName ,M.Email, BTitle ,Loan_Date, Due_Date,LStatus
from Loan L JOIN Members M ON L.FM_Id = M.M_Id
JOIN Book B ON L.FB_Id = B.B_Id
WHERE L.LStatus IN ('Issued', 'Overdue');
```

FName	Email	BTitle	Loan_Date	Due_Date	LStatus
Ahmed Ali	ahmed@gmail.com	Arab History	2025-12-10	2025-12-20	Issued
Fatima Hassan	fatima@gmail.com	Desert Stories	2025-12-11	2025-12-21	Issued
Mohamed Said	mohamed@gmail.com	Science Basics	2025-12-12	2025-12-22	Issued
Aisha Omar	aisha@gmail.com	Kids Tales	2025-12-13	2025-12-23	Issued
Khalid Salem	khalid@gmail.com	Islamic Culture	2025-12-14	2025-12-24	Issued

```
--Section 2: Aggregate Functions and Grouping

--8. Monthly Loan Statistics
SELECT
    DATENAME(MONTH, Loan_Date) AS MonthName,
    COUNT(*) AS TotalLoans,
    SUM(CASE WHEN LStatus = 'Returned' THEN 1 ELSE 0 END) AS Returned,
    SUM(CASE WHEN LStatus IN ('Issued','Overdue') THEN 1 ELSE 0 END) AS Active
FROM Loan
WHERE YEAR(Loan_Date) = YEAR(GETDATE())
GROUP BY DATENAME(MONTH, Loan_Date), MONTH(Loan_Date)
ORDER BY MONTH(Loan_Date);
```

MonthName	TotalLoans	Returned	Active
December	5	0	5

--Section 3: Views Creation

--13. vw_CurrentLoans

CREATE VIEW vw_CurrentLoans AS

SELECT

M.FName,

B.BTitle,

L.Loan_Date,

L.Due_Date,

L.LStatus,

DATEDIFF(DAY, GETDATE(), L.Due_Date) AS DaysToDue

FROM Loan L

JOIN Members M ON L.FM_Id = M.M_Id

JOIN Book B ON L.FB_Id = B.B_Id

WHERE L.LStatus IN ('Issued', 'Overdue');

1 %

Results Messages

	FName	BTitle	Loan_Date	Due_Date	LStatus	DaysToDue
1	Ahmed Ali	Arab History	2025-12-10	2025-12-20	Issued	-11
2	Fatima Hassan	Desert Stories	2025-12-11	2025-12-21	Issued	-10
3	Mohamed Said	Science Basics	2025-12-12	2025-12-22	Issued	-9
4	Aisha Omar	Kids Tales	2025-12-13	2025-12-23	Issued	-8
5	Khalid Salem	Islamic Culture	2025-12-14	2025-12-24	Issued	-7

--10. Library Performance Comparison

SELECT

Lib.LName,

COUNT(DISTINCT B.B_Id) AS TotalBooks,

COUNT(DISTINCT L.FM_Id) AS ActiveMembers,

SUM(P.Amount) AS Revenue,

CAST(COUNT(B.B_Id) AS FLOAT) / NULLIF(COUNT(DISTINCT L.FM_Id),0) AS AvgBooksPerMember

FROM Libraries Lib

LEFT JOIN Book B ON Lib.Library_Id = B.FLibrary_Id

LEFT JOIN Loan L ON B.B_Id = L.FB_Id

LEFT JOIN Payment P ON L.Loan_ID = P.FLoan_ID AND P.Method='Cash'

GROUP BY Lib.LName;

%

Results Messages

	LName	TotalBooks	ActiveMembers	Revenue	AvgBooksPerMember
	Al Amal Library	12	0	NULL	NULL
	Al Hikma Library	12	0	NULL	NULL
	Al Noor Library	16	5	19.00	3.2

```
--4. Staff Performance Overview
SELECT
    Lib.LName,
    S.FName,
    S.Position,
    COUNT(B.B_Id) AS BooksManaged
FROM Staff S
JOIN Libraries Lib ON S.FLibrary_Id = Lib.Library_Id
LEFT JOIN Book B ON Lib.Library_Id = B.FLibrary_Id
GROUP BY Lib.LName, S.FName, S.Position;
```

4 %

Results Messages

	LName	FName	Position	BooksManaged
1	Al Amal Library	Rania Ahmed	Librarian	12
2	Al Amal Library	Salim Omar	Assistant	12
3	Al Hikma Library	Fahad Khalid	Manager	12
4	Al Hikma Library	Huda Salem	Assistant	12
5	Al Noor Library	Hassan Ali	Manager	16
6	Al Noor Library	Mona Said	Librarian	16